University of Constantine 2 -- Abdelhamid Mehri

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Semester1

Advanced Data bases Techniques

– Tutorial 3–

Chapter 2 : Indexing in Databases

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Objectives of Tutorial 3:

1. Master the concept of indexing.
2. Master the metrics used to calculate data access cost

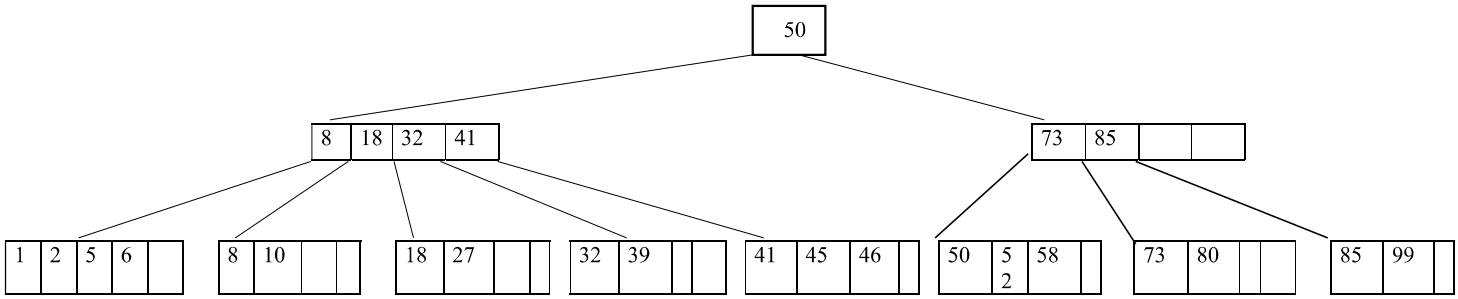
Exercice 1

Given the following set of key values: 2, 3, 5, 7, 11, 17, 19, 23, 29, 31. We want to construct the corresponding B+ tree, assuming the tree is initially empty and the values are inserted in ascending order.

Question 1: Construct the B+ tree knowing that an internal node can have at most M = 4 children
Question 2: Construct the B+ tree knowing that M = 6
Question 3: What can we deduce about the relationship between the order (i.e., the number of children per internal node) and the height of a B+ tree?

Exercice 2

Soit l'arbre B+ d'ordre 2 ci-dessous



1. What is the state of the tree after inserting the value **9**?
2. What is the state of the tree after inserting the value **3** into the initial tree?
3. What is the state of the tree after deleting the value **8** from the initial tree?
4. What is the state of the tree after inserting the value **46**, followed by deleting **85** from the initial tree?
5. What is the maximum number of elements that can be added to this tree without increasing its height? Justify your answer.

Exercice 3

We consider a database containing the relation **Student (name, first_name, School)**. We want to index the students on the attribute School. The relation contains 12 different schools, whose codes are: 1, 4, 5, 7, 10, 12, 13, 15, 16, 19, 21, and 32. To do this, we build an extendible hashing structure containing one entry for each of the 12 values. The global depth of the directory is 2, and the local depth of the index (buckets) is also 2. Therefore, each bucket can hold between 1 and 4 values.

Questions :

1. Draw the index after inserting the 12 values. Represent the directory with the names of the buckets it contains and its global depth. Represent each bucket with the values it contains and its local depth.
2. Draw the index after inserting the value 11.
3. Draw the index after inserting the value 20.
4. Draw the index after inserting the value 27.
5. What is the necessary condition to divide the global depth by 2 after deleting a value from the index?